

Wireless Security Threats and Defenses — Guided Notes

Unit 4.3.1 | CompTIA Network+ N10-009 Obj. 4.4 | N10-008 4.4

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

WEP is broken because its 24-bit _____ repeats on busy networks within hours. Once an attacker collects enough repeated IVs, they can mathematically recover the _____. The correct action when WEP is found in production is to _____.

Question 2

WPA replaced WEP's broken encryption with _____ (TKIP), which was designed to run on existing hardware. WPA2 replaced TKIP with _____ using AES encryption. Why was TKIP considered a stopgap rather than a long-term solution?

Question 3

WPA2-Personal uses a _____ shared by all devices. WPA2-Enterprise uses individual credentials authenticated via a _____ server. Why does Personal mode fail in organizational environments?

Question 4

WPA3-Personal replaces PSK with _____ (SAE), which provides _____ secrecy — meaning previously captured traffic cannot be decrypted even if the passphrase is later compromised.

Question 5

802.11w Protected Management Frames (PMF) is mandatory in WPA3. It encrypts and authenticates _____ frames, which defeats _____ attacks that force clients off a legitimate AP.

Question 6

A WIDS passively _____ the RF environment for rogue APs and unauthorized activity. A WIPS adds the ability to actively _____ rogue devices by sending deauthentication frames to clients connected

to them.

Question 7

A captive portal intercepts _____ traffic and redirects unauthenticated clients to a login page. Name two security weaknesses of captive portals that make them unsuitable as a primary security control.

Question 8 — Real World Application

REAL WORLD: A school deploys WPA2-Personal for its staff SSID. Within a week, the passphrase appears on a student group chat. IT changes the password, and it leaks again within three days. Identify the root architectural problem and describe the correct solution.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/4-3-1-wireless-security-threats-and-defenses/4-3-1-wireless-security-threats-and-defenses-answer-key.pdf
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